

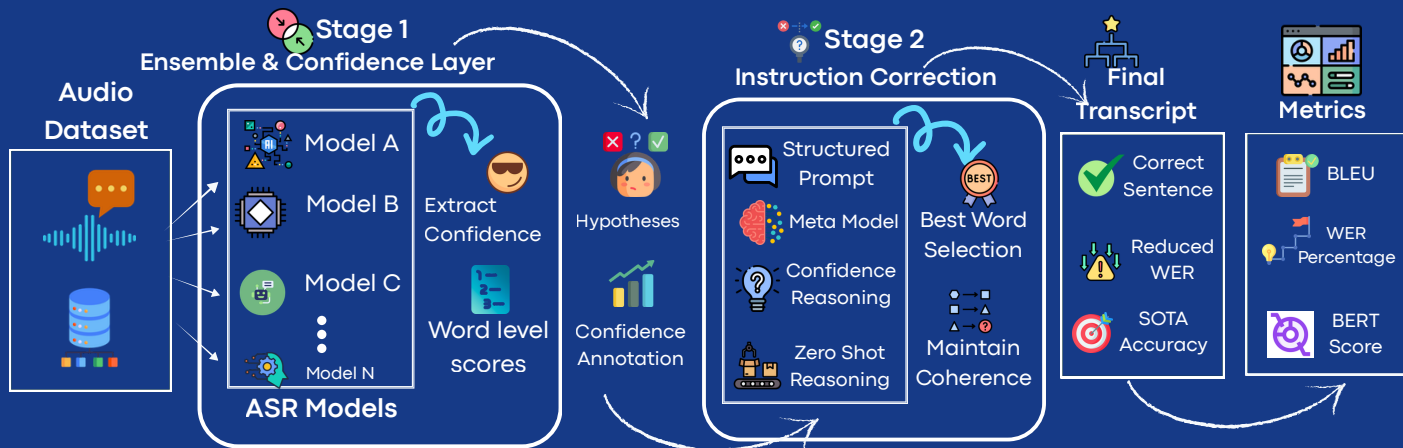
CORAL

CONSENSUS-BASED REFINEMENT AND LEARNING



Novel AI architecture combining multi-model ASR ensembles, confidence scoring, & instruction-tuned LLMs transforming Urdu speech recognition.

ARCHITECTURE



1. Multi-ASR Decoding

Run multiple ASR models in parallel to get word tokens, timestamps, and confidence scores for every token.

2. Confidence Normalization

Convert raw model scores to a uniform 0-1 scale and calibrate them using held-out data.

3. LLM Fusion

Provide aligned alternatives with their confidences to an instruction-tuned LLM. The LLM outputs a single transcript.

4. Comparison with Baselines

Evaluate against ROVER (voting), confusion networks, and highest-confidence fusion.

TIMELINE

Iteration 1

(Sep - Oct 2025)

Foundation & Confidence Extraction

- Integrate ASR models
- Extract word-level confidences
- Establish baselines.

Iteration 2

(Nov - Dec 2025)

Instruction Prompt Development

- Design prompts
- Test LLM behavior
- Structured input/output parsing

Iteration 3

(Feb - Mar 2026)

End-to-End Integration

- Integrate pipeline
- Run preliminary evaluations

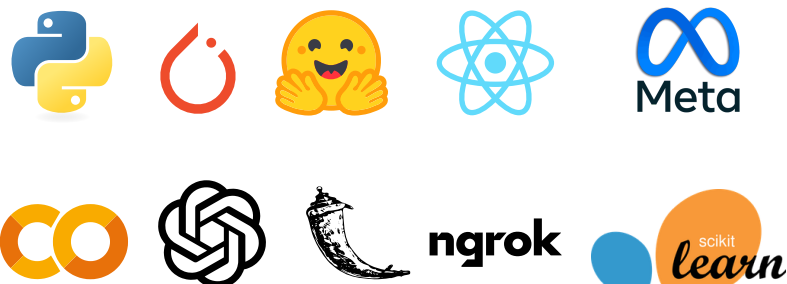
Iteration 4

(Apr - May 2026)

Optimization & Evaluation

- Calibrate
- Profile latency and optimize
- Prepare final report

TOOLS



TEAM MEMBERS

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